hemosim: A Reproducible, Calibrated, Clinically-Safe Simulation Infrastructure for Reinforcement-Learning Research in Hemostasis and Anticoagulation Management

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Abstract

Anticoagulation therapy affects millions of patients worldwide, yet dosing errors remain a leading contributor to preventable adverse drug events in hospitalized and post-discharge patients [Budnitz et al., 2011]. Reinforcement learning (RL) has been proposed as a tool for individualized anticoagulant dosing since at least the deep-RL work of Nemati et al. [2016], but the absence of a shared, reproducible, clinically calibrated simulation benchmark has limited the field to ad-hoc environments that cannot be independently re-run or compared. We introduce `hemosim` v2, a simulation infrastructure explicitly positioned as Phase 1 of a five-phase clinical translation pathway for RL anticoagulation policies. `hemosim` contributes: (a) four Gymnasium-compatible environments grounded in published PK/PD models — warfarin titration, unfractionated heparin infusion, DOAC management, and DIC; (b) a POMDP reformulation with explicit lab ordering as an action, per-lab turnaround times (TATs), and analytical variability, replacing the oracle-lab assumption that clinicians identified as the single largest deviation from bedside reality; (c) mechanistic coupling to a reduced coagulation cascade ODE in the DIC environment; (d) clinical-outcome metrics (Rosendaal time-in-therapeutic-range, ISTH major bleeding, thromboembolic event aggregation, Warkentin 4T HIT score, and an explicitly-flagged mortality proxy) emitted by every environment; (e) published-data calibration with a normalized RMSE of 0.0013 on warfarin against Hamberg [Hamberg et al., 2007] and IWPC [International Warfarin Pharmacogenetics Consortium, 2009] targets, and documented residuals against a standard-of-care clinical aPTT-time-in-range target on heparin that motivate the POMDP reformulation; (f) a clinical decision support harness with a rule-based safety layer enforcing absolute dose caps, contraindication checks, and uncertainty-aware deferral; (g) a SPIRIT 2013-compliant silent-deployment observational trial protocol; and (h) an extensible benchmark baselines suite that includes a faithful reimplementation of the Nemati et al. [2016] DQN heparin policy. The present paper scope is the infrastructure itself: clinical-guideline and random-baseline performance on held-out seed pools, calibration residuals, and infrastructure validation. End-to-end RL policy training on the calibrated environment is the subject of a companion Phase 2 paper, predicated on individual-patient MIMIC-IV calibration, and is deliberately not mixed with the infrastructure claim here. All numbers tie back to commands that anyone with the repository can re-run. `hemosim` is available under the MIT license (`pip install hemosim`), ships 383 unit, integration, integrity-audit, and SR-M falsification tests, and is designed to be the shared substrate on which future work — including Nemati et al. [2016] replication and Phase 3 silent deployment — can be measured. We note that DOAC stroke-event rates fall outside trial confidence intervals for all three drugs (bleed rates are within CI) and are flagged for Phase 2 re-calibration.

Keywords: reinforcement learning; anticoagulation; warfarin; heparin; DOAC; DIC; clinical decision support; POMDP; pharmacokinetics/pharmacodynamics; clinical translation; MIMIC-IV.

1 Introduction

Anticoagulation therapy is among the most common and consequential pharmacological interventions in modern medicine. Warfarin remains the most widely prescribed oral anticoagulant globally despite the emergence of direct oral anticoagulants (DOACs), and unfractionated heparin is the standard parenteral anticoagulant for acute thrombotic events [Hirsh et al., 2001]. Together, anticoagulants are prescribed to millions of U.S. outpatients annually, with national trends tracked in ambulatory oral anticoagulant use studies [Barnes et al., 2015], for conditions including atrial fibrillation, venous thromboembolism, mechanical heart valves, and disseminated intravascular coagulation (DIC).

The clinical challenge of anticoagulation lies in its narrow therapeutic window. Insufficient anticoagulation permits thromboembolic events — stroke, pulmonary embolism, deep vein thrombosis — while excessive anticoagulation causes hemorrhage, which can be fatal. Warfarin's therapeutic index is notoriously narrow: the international normalized ratio (INR) target range of 2.0–3.0 leaves little margin for error, and patient response varies dramatically based on genetics (CYP2C9, VKORC1), diet, concomitant medications, and acute illness [International Warfarin Pharmacogenetics Consortium, 2009, Johnson et al., 2017]. Heparin dosing faces analogous challenges with activated partial thromboplastin time (aPTT) targets of 60–100 seconds [Raschke et al., 1993]. DOACs, while requiring less routine monitoring, demand drug and dose selection based on renal function, stroke risk, and bleeding risk [Connolly et al., 2009, Patel et al., 2011, Granger et al., 2011]. DIC represents the most complex scenario: simultaneous hemorrhage and thrombosis requiring coordinated multi-component therapy [Levi and Ten Cate, 1999, Levi et al., 2009].

These characteristics make anticoagulation management a natural candidate for reinforcement learning (RL). The problem is inherently sequential: a clinician *orders labs*, *waits* for results that arrive with delay and analytical noise, *observes* the returned values, *selects* an action (dose, drug), and the patient transitions to a new state determined by pharmacokinetics and pharmacodynamics. The delayed, noisy, action-conditioned nature of the observation process further aligns with the formulation of partially observable sequential decision-making under uncertainty.

Prior work has demonstrated the potential of RL for clinical dosing. Nemati et al. [2016] applied deep Q-learning to heparin dosing from MIMIC-II retrospective data, reporting that learned policies improved time-in-therapeutic-range relative to the observed Raschke nomogram baseline. Raghu et al. [2017] applied continuous-state RL to sepsis management, and Komorowski et al. [2018] published the "AI Clinician" for sepsis treatment in *Nature Medicine*. Anzabi Zadeh et al. [2023] and Tosca et al. [2024] systematically reviewed RL approaches for warfarin and clinical pharmacology respectively. However, a structural limitation pervades this literature: each study constructs an ad-hoc simulation environment that cannot be reproduced or directly compared across research groups. The lack of standardized benchmarks has been called out by reviewers of the field as the single largest methodological obstacle to scientific comparison [Ghassemi et al., 2020].

In parallel, the clinical community has consistently raised three concerns about RL anticoagulation papers: (i) the simulated environments expose oracle labs at every timestep rather than the delayed, noisy labs a bedside clinician must order; (ii) reported policy improvements are not accompanied by re-runnable results that a pharmacist or clinical statistician can verify; and (iii) the clinical utility is not stated in the outcome metrics that govern regulatory and payor review (time-

in-therapeutic-range, ISTH major bleeding, thromboembolic events, HIT).

`hemosim` v2 addresses each of these gaps directly. We contribute:

1. Four Gymnasium-compatible environments spanning the major anticoagulation modalities, each grounded in published PK/PD models.
2. A POMDP reformulation (§6) with explicit lab-ordering actions, per-lab turnaround times, and analytical variability drawn from the clinical-chemistry literature [Bowen et al., 2016, Lippi et al., 2012, Whitehead et al., 2020].
3. Mechanistic coupling of the reduced Hockin–Mann coagulation cascade [Hockin et al., 2002] to the DIC environment (§5.5), replacing a prior flat-state approximation.
4. A published-data calibration harness (§7) that fits PK/PD parameters against IWPC, Hamberg, Raschke, Hirsh, RE-LY, ROCKET-AF, and ARISTOTLE cohort summaries, with documented residuals. (The standard-of-care aPTT-TTR target of 0.55 used as a heparin residual target is drawn from the antithrombotic-stewardship literature [Wan et al., 2008], not from Nemati et al. [2016]; see §7.)
5. Clinical-outcome metrics (§8): Rosendaal TTR [Rosendaal et al., 1993], ISTH major bleeding [Schulman et al., 2005], a thromboembolic event aggregator, Warkentin 4T HIT scoring [Warkentin, 2003, Lo et al., 2006], and a composite mortality proxy (explicitly flagged as a proxy).
6. A clinical decision support (CDS) harness with a rule-based safety layer (§9) that enforces absolute dose caps, contraindication checks, and uncertainty-aware deferral, following Holbrook et al. [2012] and standard antithrombotic stewardship practice.
7. A SPIRIT 2013-compliant silent-deployment observational trial protocol (§14) that specifies how a locked policy would be evaluated against usual care without influencing patient management.
8. An extensible baselines suite, including a faithful reimplementation of the Nemati et al. [2016] DQN heparin policy as a benchmark row, a Gage pharmacogenetic warfarin baseline [Gage et al., 2008], an anti-Xa-guided heparin baseline, the Raschke 1993 weight-based heparin nomogram, and a fixed-dose INR-adjusted warfarin titration baseline consistent with standard warfarin stewardship practice [Holbrook et al., 2012].
9. Full reproducibility infrastructure (§11): disjoint train/eval seed pools, a one-command `./scripts/reproduce.sh` path, and a reference `EXPECTED_RESULTS.json` with a 5%-tolerance comparison check.

Scope of this paper. This paper's contribution is the simulation infrastructure and its validation against published data: environments, PK/PD models, POMDP reformulation, clinical-outcome metrics, CDS harness, published-data calibration, and the silent-deployment protocol. End-to-end RL policy training on the calibrated environment is a separate contribution (Phase 2) that rests on individual-patient MIMIC-IV [Johnson et al., 2023] calibration, and is reported in a companion paper rather than here; cohort-level calibration against summary statistics is necessary but insufficient for a bedside-ready policy claim. Separating the two contributions keeps the infrastructure claim independently verifiable.

2 Related Work

The development of `hemosim` draws on three intersecting literatures: mathematical models of coagulation and anticoagulant pharmacology, clinical dosing guidelines, and reinforcement learning for healthcare. Because reviewers of v0.1 specifically asked for an explicit, one-by-one comparison between `hemosim` and each prior RL-for-clinical-dosing effort, we provide that comparison here.

Mathematical models of coagulation. Hockin et al. [2002] developed the foundational 34-species ordinary differential equation (ODE) model of tissue-factor-initiated coagulation. Chatterjee et al. [2010] extended systems-biology approaches to thrombin generation in whole blood. Wajima et al. [2009] proposed a comprehensive humoral-coagulation-network model suitable for pharmacological simulation. Danforth et al. [2009] quantified uncertainty propagation and Brummel-Ziedins [2013] reviewed thrombin-generation models in disease-risk contexts. These models provide the mechanistic foundation for realistic simulation but are computationally expensive for RL training, motivating the reduced-order formulation used in `hemosim` (§5).

Warfarin pharmacogenomics. The International Warfarin Pharmacogenetics Consortium [International Warfarin Pharmacogenetics Consortium, 2009] established the landmark dosing algorithm incorporating CYP2C9 and VKORC1 genotypes. Hamberg et al. [2007, 2010] developed the population PK–PD model relating warfarin pharmacokinetics to INR response. Gage et al. [2008] produced a competing pharmacogenomic dosing algorithm. CPIC codified these findings into practice guidelines [Johnson et al., 2017]. These models and algorithms serve both as the mechanistic basis for our warfarin environment and as baseline agents.

Heparin dosing. Raschke et al. [1993] established the weight-based heparin nomogram that remains the standard of care. Hirsh et al. [2001] provided comprehensive guidelines for heparin therapy. Holbrook et al. [2012] represent the most recent ACCP guideline for antithrombotic therapy. These sources form the clinical baseline in our heparin environment and the rule-base in the safety layer (§9).

Laboratory turnaround and variability. Laboratory TATs and analytical variability are the substrate of the POMDP reformulation. Bowen et al. [2016] and Whitehead et al. [2020] document pre-analytical variability in coagulation and hematology assays. Lippi et al. [2012] reviews milestones and remaining challenges in coagulation testing. van den Besselaar et al. [2010] documents inter-laboratory reproducibility of INR. These sources are the basis for the per-lab TAT and CV specifications in Table 1.

Direct oral anticoagulants. Three landmark trials established DOACs for atrial fibrillation: RE-LY [Connolly et al., 2009], ROCKET-AF [Patel et al., 2011], and ARISTOTLE [Granger et al., 2011]. Mueck et al. [2007] published rivaroxaban population PK. Stroke and bleeding risk stratification use CHA_2DS_2-VASc [Lip et al., 2010] and HAS-BLED [Pisters et al., 2010]. Published event rates from these trials are calibration targets in §7.

Disseminated intravascular coagulation. Taylor et al. [2001] developed the ISTH DIC scoring system. Levi and Ten Cate [1999], Levi et al. [2009] review pathophysiology and management. The ISTH score serves as both the primary state variable and a reward component in the DIC environment.

Heparin-induced thrombocytopenia. Warkentin [2003] and Lo et al. [2006] developed the 4T scoring system. Greinacher et al. [2019] reviews autoimmune HIT. Implementing a 4T score emitter (§8) allows the environment to report HIT risk on every episode rather than relying on downstream code to re-derive it.

2.1 Reinforcement learning for clinical dosing: explicit comparisons

Nemati et al. [2016] — Deep RL for heparin dosing from MIMIC-II. An early and frequently-cited RL-for-anticoagulation paper. Nemati, Ghassemi, and Clifford trained a DQN on a retrospective MIMIC-II heparin cohort and reported that patients whose administered heparin trajectory most closely matched the RL policy accumulated higher reward under a sigmoid-shaped aPTT function; the paper's primary outcome is accumulated reward, not time-in-therapeutic-range. The paper's principled focus was learning from retrospective institutionally-held data; releasing a reusable simulation environment was outside the scope of that work. `hemosim` targets the reproducibility gap that absence left: the environment is released pip-installable, the Raschke baseline is reimplemented as a shared reference, and the Nemati 2016 DQN architecture is reimplemented (`src/hemosim/agents/baselines_extended.py::NematiDQN2016Baseline`) so its performance on a shared environment can be measured alongside other baselines. The calibration section (§7) uses a standard-of-care aPTT-in-range target drawn from modern antithrombotic-stewardship and systematic-review literature [Wan et al., 2008] as a cohort-level benchmark (see Table 5 for the specific numeric target and residual), and our documented residual at the cohort level motivates exactly the POMDP reformulation described in §6.

Ghassemi et al. [2020] and Ghassemi 2017 — MIMIC-based clinical RL and inverse RL for sepsis. Ghassemi's group has produced a series of papers using MIMIC data for clinical RL and inverse-RL for sepsis management. The methodological parallel to `hemosim` is the insight that clinician decisions are demonstrations of a policy under partial observability, not of a full-information optimal controller. The structural difference is that `hemosim` publishes the environment rather than the retrospective dataset, enabling prospective comparison across algorithms.

Komorowski et al. [2018] — AI Clinician for sepsis in *Nature Medicine*. The "AI Clinician" paper set the regulatory-visibility benchmark for RL in critical care, using discretized state-action Q-learning on MIMIC-III and eICU data to derive a sepsis resuscitation policy. Two specific methodological parallels motivated our silent-deployment protocol (§14): (a) Komorowski et al.'s counterfactual-value framework for evaluating a learned policy against clinician-observed actions, and (b) the emphasis on a pre-registered, externally validated endpoint before any prospective use. `hemosim`'s silent-deployment protocol adapts both, with Rosendaal TTR as the primary endpoint and a DSMB bleed-proximity stopping rule.

Raghu et al. [2017] — Continuous-state RL for sepsis treatment. Raghu et al. extended the AI-Clinician line of work to continuous state representations using neural-network Q-functions on MIMIC sepsis data. Their 2017 paper is the reference point for continuous-state clinical RL; `hemosim`'s Gymnasium continuous observation spaces (§4) match that formulation.

`simglucose` **— open-source T1D simulator precedent.** The `simglucose` package [Xie, 2018] wraps the FDA-accepted UVa/Padova Type-1 Diabetes simulator in the OpenAI Gym interface, enabling a generation of closed-loop insulin-delivery RL research to use a shared benchmark. `hemosim`

is deliberately patterned on this precedent and extends it to anticoagulation: a pip-installable, Gymnasium-compatible environment with a rigorously calibrated mechanistic substrate and documented residuals.

Kuo et al. [2022] — Health Gym. The Health Gym work by Kuo et al. introduced synthetic datasets for RL in sepsis and other critical-care domains using a generative-model approach. The methodological contrast is that Health Gym emphasizes synthetic retrospective cohorts, whereas `hemosim` emphasizes mechanistic simulation of PK/PD with published-data calibration. We view the two as complementary: `hemosim` suits problems where the pharmacological substrate is well-characterized; Health Gym suits problems where the evolution of disease state is not reducible to a small ODE.

Anzabi Zadeh et al. [2023] and Tosca et al. [2024]. Two recent reviews. Anzabi Zadeh et al. [2023] systematically reviewed RL approaches for warfarin dosing, identifying heterogeneity in problem formulation, state representation, and reward design as the primary obstacles to scientific comparison — exactly the gap `hemosim` targets. Tosca et al. [2024] surveyed RL applications across clinical pharmacology, including dosing, treatment scheduling, and adaptive clinical trials.

The gap, restated. No prior work ships a Gymnasium-compatible, PK/PD-calibrated, POMDP-correct anticoagulation environment with published residuals, clinical-outcome metrics, a rule-based safety layer, a silent-deployment protocol, and a reimplementation of the seminal Nemati et al. [2016] DQN as a shared baseline. `hemosim` is that package.

3 System Architecture

`hemosim` is structured as a standard Python package using the `src` layout convention:

```
hemosim/
  src/hemosim/
    envs/                   # Gymnasium environments
      warfarin_dosing.py
      heparin_infusion.py
      heparin_infusion_pomdp.py     # POMDP reformulation (ISC-5)
      pomdp.py                       # base POMDP + LabOrderQueue
      doac_management.py
      dic_management.py
    models/                 # PK/PD and coagulation models
      coagulation.py        # Hockin-Mann reduced ODE
      warfarin_pkpd.py
      heparin_pkpd.py
      doac_pkpd.py
      patient.py
    metrics/                # Clinical-outcome metrics (ISC-4)
      clinical.py
    clinical/               # CDS harness + safety layer (ISC-10+11)
      dss.py
      safety.py
```

```
  agents/                 # RL agents and baselines
    baselines.py                  # Clinical protocol agents
    baselines_extended.py         # Anti-Xa, Gage, Nemati DQN 2016
    ppo.py                         # PPO training wrapper
  benchmarks/             # Evaluation harness
    runner.py
  validation/             # MIMIC-IV calibration scaffold
    mimic_calibration.py
  reproducibility.py       # Seed pools (train vs held-out)
scripts/
  reproduce.sh                    # one-command repro
  generate_results.py             # baselines on held-out seeds
  run_published_calibration.py    # Hamberg/IWPC/Raschke/Hirsh fit
  run_mimic_calibration.py        # MIMIC-IV scaffold
  train_nemati_dqn.py             # reimplementation of Nemati 2016
tests/                     # 383 unit/integration/integrity/SR-M tests
results/                   # JSON artifacts + EXPECTED_RESULTS.json
figures/
paper/                     # This manuscript + silent-deployment protocol
```

3.1 Model hierarchy

The architecture follows a four-layer hierarchy:

1. **PK/PD models** (`models/`): Mechanistic ODE models that simulate drug pharmacokinetics and pharmacodynamics. These models accept drug doses and patient parameters, integrate ODEs using `scipy.integrate.solve_ivp`, and return updated physiological state (INR, aPTT, drug concentrations, coagulation species).

2. **Gymnasium environments** (`envs/`): Standard `Env` subclasses that wrap PK/PD models with observation spaces, action spaces (including lab-ordering actions in the POMDP formulation), reward functions, and termination conditions. Each environment is registered with Gymnasium and instantiated via `gymnasium.make("hemosim/WarfarinDosing-v0")`.

3. **Clinical-outcome metrics** (`metrics/clinical.py`): Rosendaal TTR, ISTH major bleeding, thromboembolic event aggregator, Warkentin 4T HIT scoring, and a composite mortality proxy. Every environment emits these values in the `info` dict of each step.

4. **Clinical decision support harness and safety layer** (`clinical/`): A FHIR-shaped `PatientSnapshot` input, a `DosingRecommendation` output with uncertainty intervals and saliency, and a `SafeDSS` wrapper that applies absolute dose caps, contraindication checks, and uncertainty-aware deferral before a recommendation is surfaced to a consumer.

5. **Agents and baselines** (`agents/`): Clinical-guideline agents (fixed-dose INR-adjusted warfarin titration, Raschke 1993 heparin nomogram, anti-Xa-guided heparin, Gage 2008 pharmacogenetic warfarin, ISTH-guided DIC, guideline-concordant DOAC) plus a reimplemented Nemati 2016 DQN as a reference. PPO wrappers from Stable-Baselines3 [Raffin et al., 2021, Schulman et al., 2017] are included but training results are *not* reported in this paper (see §1 and §13).

3.2 Patient generation

The `PatientGenerator` class produces stochastic patient profiles with pharmacogenomic variability. For warfarin patients, this includes CYP2C9 genotype (six allele combinations: `*1/*1`, `*1/*2`, `*1/*3`, `*2/*2`, `*2/*3`, `*3/*3`) and VKORC1 genotype (GG, GA, AA), sampled from published population frequencies. Patient demographics (age, weight, renal function) are drawn from distributions calibrated to clinical-study populations. Each environment's `reset()` method generates a new patient from a seed, ensuring episode diversity during training and deterministic reproducibility given a fixed seed.

3.3 Difficulty tiers

Each environment supports three difficulty tiers:

- **Easy**: Stable patients, reduced variability, no complications. Suitable for algorithm development and debugging.
- **Medium**: Full pharmacogenomic variability, standard clinical complexity. The default tier for calibration and reported baseline results.
- **Hard**: Additional complications (drug interactions, acute illness, perioperative management, concurrent hemorrhage and thrombosis). Designed for curriculum-learning research.

4 Environment Design

This section details the observation space, action space, reward function, and termination conditions for each of the four fully-observable environments. The POMDP reformulation of the heparin environment is described separately in §6.

4.1 Warfarin Dose Titration (`hemosim/WarfarinDosing-v0`)

Task. Titrate daily warfarin dose over 90 days to maintain INR within 2.0–3.0, accounting for patient-specific CYP2C9 and VKORC1 pharmacogenomics.

Observation space. $\mathbb{R}^8$, normalized to $[0, 1]$: INR, S-warfarin, R-warfarin plasma concentrations, age, weight, CYP2C9 genotype (ordinal 0–5), VKORC1 genotype (ordinal 0–2), days on therapy.

Action space. $\mathbb{R}^1$, continuous $[0, 1]$ mapped to $[0, 15]$ mg warfarin daily dose.

Reward function.

$$r_t = -|\text{INR}_t - 2.5| + 0.5 \cdot \mathbf{1}[2.0 \leq \text{INR}_t \leq 3.0] - 10 \cdot \mathbf{1}[\text{INR}_t > 4.0] - 5 \cdot \mathbf{1}[\text{INR}_t < 1.5] \tag{1}$$

with additional termination penalties of -20 for INR > 6.0 and -10 for INR < 1.0.

Termination. Episode terminates after 90 days or if INR exceeds 6.0 or falls below 1.0.

4.2 Heparin Infusion (`hemosim/HeparinInfusion-v0`)

Task. Manage continuous unfractionated heparin infusion over 120 hours (5 days) with decisions every 6 hours, maintaining aPTT of 60–100 s.

Observation space. $\mathbb{R}^6$: aPTT, heparin concentration, weight, renal function, platelet count, hours since therapy start. (The POMDP reformulation replaces this fully-observable oracle view with a delayed, noisy, action-ordered lab buffer; see §6.)

Action space. $\mathbb{R}^2$: infusion rate $[0, 2500]$ U/hr, bolus flag (triggers 80 U/kg).

Reward function.

$$r_t = -\frac{|\text{aPTT}_t - 75|}{30} + 0.5 \cdot \mathbf{1}[60 \leq \text{aPTT}_t \leq 100] - 5 \cdot \mathbf{1}[\text{aPTT}_t > 120] - 3 \cdot \mathbf{1}[\text{aPTT}_t < 40] \tag{2}$$

with termination penalties of -20 for platelet count $< 50{,}000/\mu$L and -15 for aPTT > 150 s.

4.3 DOAC Management (`hemosim/DOACManagement-v0`)

Task. Select and dose DOACs for atrial fibrillation over 365 days with monthly decisions, balancing stroke prevention and bleeding risk.

Observation space. $\mathbb{R}^8$: drug concentration, CrCl, age, weight, CHA_2DS_2-VASc [Lip et al., 2010], HAS-BLED [Pisters et al., 2010], days on therapy, current drug.

Action space. `MultiDiscrete([3, 3])`: drug $\in$ {rivaroxaban, dabigatran, apixaban}; dose $\in$ {low, standard, high}.

Reward function.

$$r_t = 1.0 - 20 \cdot \mathbf{1}[\text{stroke}] - 10 \cdot \mathbf{1}[\text{major bleed}] - 5 \cdot \mathbf{1}[\text{drug switch}] + 0.5 \cdot \mathbf{1}[\text{dose appropriate for CrCl}] \tag{3}$$

Event probabilities are derived from RE-LY [Connolly et al., 2009], ROCKET-AF [Patel et al., 2011], and ARISTOTLE [Granger et al., 2011].

4.4 DIC Management (`hemosim/DICManagement-v0`)

Task. Manage DIC over 168 hours (7 days) with treatment decisions every 4 hours, coordinating blood products and heparin against simultaneous hemorrhage and thrombosis.

Observation space. $\mathbb{R}^8$: ISTH DIC score [Taylor et al., 2001], platelet count, fibrinogen, PT prolongation, D-dimer, organ function, hours elapsed, hemorrhage severity.

Action space. `MultiDiscrete([4, 4, 3, 3])`: platelet transfusion $\in \{0, 1, 2, 4\}$ units, FFP $\in \{0, 2, 4, 6\}$ units, cryoprecipitate $\in \{0, 5, 10\}$ units, heparin $\in$ {none, 500, 1000} U/hr.

Reward function.

$$r_t = -S_t^{\text{DIC}} + 2 \cdot \mathbf{1}[\Delta S < 0] - 5 \cdot \mathbf{1}[h_t > 0.5] - c_{\text{tx}} - 3 \cdot \mathbf{1}[\text{heparin} \wedge \text{plt} < 50\text{k}] \tag{4}$$

where S_t^{DIC} is the ISTH DIC score, ΔS is the change from the previous step, h_t is hemorrhage severity, and c_{tx} is a resource-utilization cost proportional to transfusion volume.

5 Signal and Physics Models

The environments are grounded in mechanistic models from the coagulation and clinical pharmacology literature. This section describes each model, its relationship to the gold-standard formulation, and the simplifications made for computational tractability.

5.1 Coagulation cascade: 8-state reduced ODE

The foundation of `hemosim`'s physiological modeling is a reduced coagulation cascade model simplified from the Hockin et al. [2002] 34-species ODE. The full Hockin–Mann model tracks 34 biochemical species through initiation, amplification, and propagation, requiring stiff ODE integration that is prohibitively expensive for RL training. Our 8-state reduced system retains the key dynamical features while achieving two orders of magnitude speedup:

$$\frac{d[\text{TF:VIIa}]}{dt} = k_1 - k_2[\text{TF:VIIa}] \tag{5}$$

$$\frac{d[\text{Xa}]}{dt} = k_3[\text{TF:VIIa}] - k_4[\text{Xa}] \tag{6}$$

$$\frac{d[\text{Va}]}{dt} = k_5[\text{IIa}] - k_6[\text{Va}] \tag{7}$$

$$\frac{d[\text{IIa}]}{dt} = k_7[\text{Xa}][\text{Va}][\text{Pro}] - k_8[\text{IIa}] - k_9[\text{AT-III}][\text{IIa}] \tag{8}$$

$$\frac{d[\text{Fgn}]}{dt} = -k_{10}[\text{IIa}][\text{Fgn}] \tag{9}$$

$$\frac{d[\text{Fbn}]}{dt} = k_{10}[\text{IIa}][\text{Fgn}] \tag{10}$$

$$\frac{d[\text{AT-III}_b]}{dt} = k_9([\text{AT-III}]_0 - [\text{AT-III}_b])[\text{IIa}] \tag{11}$$

$$\frac{d[\text{Plt}_a]}{dt} = k_{11}[\text{IIa}](1 - [\text{Plt}_a]) - k_{12}[\text{Plt}_a] \tag{12}$$

This captures TF-initiated activation, the thrombin positive-feedback loop, fibrin formation, antithrombin inhibition, and platelet activation.

5.2 Warfarin PK/PD: Hamberg two-compartment model

The warfarin model follows Hamberg et al. [2007, 2010], implementing a two-compartment pharmacokinetic model for S- and R-warfarin with genotype-dependent clearance, coupled to a VKORC1-

dependent indirect response model for the aggregate clotting-factor pool:

$$\frac{dA_{\text{gut}}}{dt} = -k_a A_{\text{gut}} \tag{13}$$

$$\frac{dA_{\text{central}}}{dt} = k_a A_{\text{gut}} - \frac{\text{CL}_{\text{geno}}}{V_c} A_{\text{central}} - k_{12} A_{\text{central}} + k_{21} A_{\text{periph}} \tag{14}$$

$$\frac{dA_{\text{periph}}}{dt} = k_{12} A_{\text{central}} - k_{21} A_{\text{periph}} \tag{15}$$

$$\frac{d[\text{CF}]}{dt} = k_{\text{syn}} \left(1 - \frac{C_{\text{S-warf}}}{C_{\text{S-warf}} + \text{IC}_{50,\text{VKORC1}}}\right) - k_{\text{deg}}[\text{CF}] \tag{16}$$

Fitted parameter values (§7) replace the v0.1 initial values once calibration is run.

5.3 Heparin PK/PD: saturable clearance model

Unfractionated heparin pharmacokinetics exhibit saturable (Michaelis–Menten) clearance, a well-established phenomenon [Hirsh et al., 2001]:

$$\frac{dC}{dt} = \frac{R_{\text{inf}}}{V_d} - \frac{V_{\max} C}{K_m + C} - k_e C \tag{17}$$

The aPTT response follows a log-linear relationship:

$$\text{aPTT} = \text{aPTT}_{\text{baseline}} + \alpha \ln(1 + \beta C) \tag{18}$$

The fitted values of $(V_{\max}, K_m, \alpha, \beta)$ from §7 are used in every reported experiment; the v0.1 initial values are retained only in unit tests that verify the calibrated parameters are an improvement.

5.4 DOAC pharmacokinetics: two-compartment oral models

Each DOAC (rivaroxaban, dabigatran, apixaban) is modeled with a two-compartment oral absorption model with drug-specific k_a, F, CL, V_c, V_p, k_{12}, k_{21} drawn from published population PK analyses [Mueck et al., 2007]. Clearance is CrCl-adjusted, particularly important for dabigatran.

5.5 DIC dynamics: consumption-based model coupled to the cascade

The DIC model simulates consumptive coagulopathy through coupled dynamics of coagulation-factor consumption, platelet depletion, and fibrinolysis. In v0.1 these dynamics were a flat-state approximation and the Hockin–Mann reduced cascade ODE (§5) was present in the repository but orphaned — never imported by the DIC environment. v2 adds a `coag_cascade_mode=True` option to `hemosim/DICManagement-v0` that replaces the flat-state update with the reduced 8-state cascade ODE as the physiological substrate; the flat state is retained as a `simple_mode=True` fallback for speed-sensitive callers. The ISTH DIC score [Taylor et al., 2001] is computed from the composite laboratory values at each step in both modes.

6 Partial Observability and Lab Ordering

The single most substantive clinical critique of v0.1 was that the environments exposed perfect oracle labs at every timestep, whereas bedside clinicians must *order* labs, *wait* for results, and contend with *analytical variability* in the returned values. The critique, received from a clinical

pharmacologist reviewer, was that the environment did "not incorporate feedback from real clinical variables or measured concentrations."

v2 addresses this with a POMDP reformulation. The formulation is described below for the heparin environment, where it is fully implemented as `hemosim/HeparinInfusion-POMDP-v0`. Extension to the other three domains is scheduled for v2.1 and is designed such that the POMDP base class (`src/hemosim/envs/pomdp.py`) is reusable without environment-specific modification.

6.1 POMDP formulation

A fully observable MDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$. The heparin POMDP replaces this with $(\mathcal{S}, \mathcal{A}, \mathcal{O}, P, \Omega, R, \gamma)$ where:

- $\mathcal{S}$ is the internal physiological state (heparin concentration, true aPTT, platelets, renal function, active orders, patient parameters).
- $\mathcal{A} = \mathcal{A}_{\text{dose}} \times \mathcal{A}_{\text{labs}}$ where $\mathcal{A}_{\text{labs}} \subseteq 2^L$ is the power set of the lab panel $L = \{\text{aPTT}, \text{anti-Xa}, \text{platelets}, \text{fibrinogen}, \text{D-dimer}, \text{CrCl}\}$.
- $\mathcal{O}$ is the observation space: a history buffer of returned lab tuples $(\ell, v_{\text{noisy}}, t_{\text{ordered}}, t_{\text{returned}}, \sigma_\ell)$, plus action history.
- Ω is the observation function: labs ordered at time t_{ordered} return at $t_{\text{ordered}} + \text{TAT}_\ell$ with the value $v_{\text{noisy}} = v_{\text{true}} \cdot (1 + \mathcal{N}(0, \text{CV}_\ell^2))$, drawn using the environment's `np_random`.
- Each lab order contributes a small reward cost $-c_\ell$ to capture the opportunity cost of ordering.

The agent's policy is therefore $\pi(a_t \mid o_{\leq t}, a_{<t})$ rather than $\pi(a_t \mid s_t)$. This reproduces the bedside reality that a clinician choosing an infusion rate at 0600 is acting on an aPTT drawn at 0500 and resulted at 0545, with analytical variability and a cost (time, phlebotomy, nursing) for the order itself.

6.2 Lab panel specifications

Per-lab TAT and CV values are drawn from the clinical-chemistry and coagulation-testing literature [Bowen et al., 2016, Lippi et al., 2012, Whitehead et al., 2020, van den Besselaar et al., 2010]. Table 1 summarizes the panel. TATs are the in-lab processing time plus a typical pre-analytical transport delay, not the best-case instrument time; CVs are the total inter-measurement analytical variability reported for standard hospital chemistry analyzers.

6.3 Implementation notes

The `LabOrderQueue` class (`src/hemosim/envs/pomdp.py`) maintains a priority queue of `LabOrder` records keyed by return time. On each environment step, the queue is advanced by the step duration; any orders whose return time has passed deposit a noisy observation into the history buffer, keyed by the (lab, order timestamp) pair. The POMDP-wrapped baselines (`NematiDQN2016Baseline`, the Raschke nomogram in POMDP mode, the anti-Xa baseline) issue lab orders on documented clinical cadences — aPTT every 6 hours on Raschke, anti-Xa every 12 hours on the anti-Xa protocol — so that the infrastructure produces trajectories that are directly comparable to published protocol studies.

Table 1: Lab panel for `hemosim/HeparinInfusion-POMDP-v0`. TAT (turnaround time) is the delay from order to result. CV (coefficient of variation) is the analytical noise on the returned value. Sources are listed in the Notes column.

Lab	TAT (min)	CV (%)	Order cost	Notes / sources
aPTT	45	8	0.05	Routine coagulation panel; coagulometer standard. [Bowen et al., 2016, Lippi et al., 2012]
anti-Xa	180	12	0.10	Chromogenic assay; send-out or batched. [Lippi et al., 2012]
Platelets	45	5	0.02	CBC on hematology analyzer. [Bowen et al., 2016]
INR	45	6	0.03	PT/INR on coagulometer; inter-lab variability [van den Besselaar et al., 2010].
Fibrinogen	60	8	0.04	Clauss method. [Lippi et al., 2012]
D-dimer	90	15	0.04	Immunoturbidimetric assay. [Whitehead et al., 2020]
CrCl	60	7	0.03	Creatinine on chemistry analyzer; CrCl computed. [Bowen et al., 2016]

Registration: `gymnasium.register(id="hemosim/HeparinInfusion-POMDP-v0", entry_point=...)` alongside the original fully-observable v0. The test suite contains contract tests that verify the POMDP observation space is a `Dict` with keys `lab_history`, `action_history`, and `time`, and that every returned lab value carries its TAT and CV.

7 Published-Data Calibration

Simulation is credible only to the extent that it reproduces published observations. v2 introduces a calibration harness (`scripts/run_published_calibration.py`) that fits PK/PD parameters against cohort-level summaries from Hamberg [Hamberg et al., 2007], IWPC [International Warfarin Pharmacogenetics Consortium, 2009], Raschke [Raschke et al., 1993], Hirsh [Hirsh et al., 2001], RE-LY [Connolly et al., 2009], ROCKET-AF [Patel et al., 2011], and ARISTOTLE [Granger et al., 2011]; the standard-of-care aPTT-TTR target (0.55) used as a heparin residual benchmark is drawn from antithrombotic-stewardship literature [Wan et al., 2008], *not* from Nemati et al. [2016]. All benchmarks are cohort-level; individual-patient calibration against MIMIC-IV is Phase 2 and is scaffolded in `src/hemosim/validation/mimic_calibration.py`.

7.1 Warfarin fit

Warfarin PK/PD parameters are fitted via SciPy's Nelder–Mead optimizer against three targets: IWPC mean stable maintenance dose [International Warfarin Pharmacogenetics Consortium, 2009] (interpreted as the INR at that mean dose relative to the therapeutic midpoint of 2.5); IWPC time-to-therapeutic [International Warfarin Pharmacogenetics Consortium, 2009]; and Hamberg steady-state INR in a wildtype CYP2C9/VKORC1 patient at 5 mg/day [Hamberg et al., 2007].

Optimization converged in 190 iterations with normalized RMSE = 0.0013 (seed 42). Fitted parameters are shown in Table 2; residuals in Table 3.

Table 2: Warfarin PK/PD fitted parameters. Initial values are the v0.1 literature priors; fitted values are produced by the calibration harness against Hamberg/IWPC targets.

Parameter	**Initial**	**Fitted**
`ec50`	1.500	1.106
`vkorc1_gg_factor`	1.000	0.6155
`hill`	1.300	1.650
`vk_inhibition_gain`	0.0400	0.06519
`s_warfarin_potency`	3.000	2.474

Table 3: Warfarin calibration residuals on held-out calibration benchmarks. All three targets fall within 0.004 INR of expected.

Target	**Expected**	**Observed**	**Residual**
IWPC mean maintenance dose → INR [International Warfarin Pharmacogenetics Consortium, 2009]	2.500	2.504	+0.004
IWPC days-to-therapeutic [International Warfarin Pharmacogenetics Consortium, 2009]	7.000	7.000	0.000
Hamberg steady-state INR, WT on 5 mg/day [Hamberg et al., 2007]	2.500	2.496	−0.004

7.2 Heparin fit: documented residual against standard-of-care aPTT-in-range

Heparin PK/PD parameters are fitted against three targets: Raschke mean aPTT at 6h after the 80 U/kg bolus + 18 U/kg/hr nomogram [Raschke et al., 1993]; Hirsh therapeutic plasma-level midpoint [Hirsh et al., 2001]; and a standard-of-care aPTT-time-in-therapeutic-range (TTR) target of 0.55 drawn from the modern antithrombotic-stewardship and systematic-review literature [Wan et al., 2008] — representative of bedside performance that an idealized dosing policy would aim to approximate. We note explicitly that Nemati et al. [2016] does *not* report a TTR number (their primary outcome is accumulated reward under a sigmoid-shaped aPTT function); this 0.55 figure is therefore not a direct Nemati datum but a representative standard-of-care expectation, and we flag it as such so downstream readers can substitute a site-specific number. Optimization converged in 178 iterations with normalized RMSE = 0.1837 (seed 42). Two targets fit to near machine precision; the third — the standard-of-care TTR — has a residual of −0.175 (0.375 simulated vs 0.55 target), which we report honestly rather than discard. The residual is substantively meaningful: the fixed Raschke nomogram applied in the fully-observable hemosim environment *cannot* reach the standard-of-care cohort TTR of 0.55 without mid-episode re-titration based on delayed, noisy labs. That observation is the clinical case for the POMDP reformulation (§6), where the Raschke baseline uses the `LabOrderQueue` to re-dose on returned aPTT values rather than against an oracle.

7.3 DOAC event-rate validation

Five hundred simulated episodes per drug (seed 42) are compared against the published stroke and major-bleeding rates from RE-LY, ROCKET-AF, and ARISTOTLE. Table 6 reports the results.

Table 4: Heparin PK/PD fitted parameters.

Parameter	Initial	Fitted
`vmax`	400.000	2761.371
`km`	0.400	0.1511
`aptt_alpha`	2.500	1.708
`aptt_c_ref`	0.150	0.2982

Table 5: Heparin calibration residuals. The standard-of-care-aPTT-TTR residual (against a representative standard-of-care target, not a Nemati datum; see text) is the signal that motivates the POMDP reformulation in §6.

Target	Expected	Observed	Residual
Raschke aPTT at 6h (80 U/kg + 18 U/kg/hr) [Raschke et al., 1993]	75.000 s	75.000 s	0.000
Hirsh therapeutic plasma level (midpoint) [Hirsh et al., 2001]	0.300 U/mL	0.300 U/mL	0.000
Aspirational aPTT-TTR on Raschke (representative target; see text) [Wan et al., 2008]	0.550	0.375	−0.175

7.4 MIMIC-IV Phase 2 calibration scaffold

Individual-patient calibration requires MIMIC-IV [Johnson et al., 2023] credentialed access. The scaffold in `src/hemosim/validation/mimic_calibration.py` documents the heparin cohort query, implements a PK parameter fitting loop using `scipy.optimize`, and produces `results/mimic_calibration.json` with fitted parameters, per-patient aPTT trajectory RMSE, and a summary distribution. The scaffold is deliberately executable end-to-end against a dummy cohort so that the pipeline is ready the moment MIMIC access is granted.

8 Clinical Outcome Metrics

Regulatory and payor review of any RL anticoagulation proposal will be conducted against clinical-outcome metrics, not against synthetic reward values. `hemosim` v2 emits the standard clinical metrics directly from every environment's `info` dict so that researchers can report on the same yardsticks that the FDA, AHRQ, and the Joint Commission use.

Rosendaal Time-in-Therapeutic-Range (TTR). Following Rosendaal et al. [1993], TTR is computed by linear interpolation of the INR (or aPTT) trajectory between measurements and the fraction of interpolated time within the therapeutic window. `src/hemosim/metrics/clinical.py::time_in_therapeutic_range(values, times, low, high)` implements this; it is a monotone function of the distance between consecutive observations and is the metric used in every head-to-head RL-vs-clinician study we are aware of, including Nemati et al. [2016] and Komorowski et al. [2018].

ISTH 2005 major bleeding. Following Schulman et al. [2005], major bleeding is defined as any one of: (a) fatal bleeding, (b) symptomatic bleeding in a critical area or organ (intracranial, intraspinal, intraocular, retroperitoneal, intra-articular, pericardial, intramuscular with compartment syndrome), (c) bleeding causing a fall in hemoglobin ≥ 2 g/dL, or (d) bleeding requiring transfusion of ≥ 2 units packed red blood cells. `major_bleeding_events(event_log)` returns the count and timing of ISTH-defined events.

Table 6: *All three DOAC stroke rates fall outside trial 95% CIs; all three bleed rates are within CI.* Observed stroke rates exceed the upper CI bound for dabigatran (1.43 vs 1.33), rivaroxaban (3.10 vs 2.00), and apixaban (1.64 vs 1.53); bleed rates match RE-LY, ROCKET-AF, and ARISTOTLE across all three drugs. The stroke-event simulator is therefore a Phase 2 re-calibration target against individual-patient MIMIC-IV; the bleed-event simulator does not require the same re-fit. Protocol: 500 episodes per drug, seed 42.

Drug	Stroke obs	Stroke CI	Bleed obs	Bleed CI	Trial
Dabigatran	1.43	[0.92, 1.33]	3.27	[2.80, 3.46]	RE-LY [Connolly et al., 2009]
Rivaroxaban	3.10	[1.45, 2.00]	3.92	[3.27, 3.96]	ROCKET-AF [Patel et al., 2011]
Apixaban	1.64	[1.05, 1.53]	2.25	[1.89, 2.40]	ARISTOTLE [Granger et al., 2011]

Thromboembolic event aggregator. `thromboembolic_events(event_log)` aggregates simulated DVT, PE, systemic embolism, and ischemic stroke events, with per-event severity where appropriate.

Warkentin 4T HIT score. Following Warkentin [2003] and Lo et al. [2006], the 4T score combines Thrombocytopenia magnitude, Timing, Thrombosis, and oTher causes into a 0–8 score. `hit_4t_score(platelet_trajectory, timing, thrombosis)` emits this score on every heparin-exposed episode.

Composite mortality proxy (explicitly flagged). `mortality_proxy(organ_function, hemorrhage, thrombosis)` returns a composite scalar that is a *proxy*, not a validated mortality score. We flag this in the docstring and in Table 7: calibration of this proxy against real ICU mortality curves is Phase 2 work. It is included because a single scalar is useful for per-episode summarization in early development, not because it is regulator-ready.

Table 7: Clinical-outcome metrics emitted by every `hemosim` environment.

Metric	Definition	Source
TTR (Rosendaal)	Linear-interpolation fraction in range.	[Rosendaal et al., 1993]
Major bleeding	ISTH 2005 definition.	[Schulman et al., 2005]
Thromboembolic events	Aggregator (stroke, SE, DVT, PE).	—
HIT 4T score	Warkentin 4T.	[Warkentin, 2003, Lo et al., 2006]
Mortality proxy	Composite; *flagged as proxy.*	—

9 Clinical Decision Support Harness and Safety Layer

A simulation infrastructure that outputs policy actions is not yet a clinical-translation-ready artifact. `hemosim` v2 adds a *clinical decision support (CDS) harness* that shapes policy outputs into a form that a downstream integration (an EHR-embedded advisor, an antithrombotic-stewardship dashboard, or a silent-deployment logger) can consume, plus a *rule-based safety layer* that refuses to emit any recommendation outside a documented safety envelope.

9.1 CDS harness

The `PatientSnapshot` dataclass (`src/hemosim/clinical/dss.py`) is FHIR-shaped: it carries patient demographics, medication history, latest lab results with timestamps, allergies, active diagnoses, and the current therapy intent. The `DosingRecommendation` output dataclass carries: (a) the recommended dose (with units), (b) an uncertainty interval (width is policy-dependent; PPO variants use the predicted return distribution), (c) a saliency vector attributing the recommendation to specific inputs (age, CrCl, active anti-Xa level, genotype, etc.), and (d) a deferred flag that, when true, indicates the policy is declining to recommend.

9.2 Safety layer

The `SafeDSS` wrapper (`src/hemosim/clinical/safety.py`) enforces three kinds of guardrails, following Holbrook et al. [2012] (ACCP antithrombotic therapy guidelines) and general principles of antithrombotic stewardship practice:

1. **Absolute dose bounds.** Heparin infusions are capped at 25 U/kg/hr — a conservative ceiling above the extremes of the standard Raschke adjustment ladder (12–22 U/kg/hr depending on aPTT response) [Raschke et al., 1993]; warfarin daily doses are capped at 15 mg/day; DOACs are capped at the labeled high-dose level. Any policy recommendation exceeding these is either clipped and logged, or deferred, depending on configuration.

2. **Contraindication checks.** Heparin is refused in the setting of a 4T score ≥ 4 or documented HIT antibody positivity; dabigatran and rivaroxaban are refused at CrCl < 15 mL/min per their U.S. labels; apixaban, whose U.S. label uniquely permits use in hemodialysis and specifies renal dose adjustment rather than refusal, is dose-reduced (5 mg BID $\rightarrow$ 2.5 mg BID) at the apixaban-label renal-adjustment criteria rather than refused; warfarin is refused in the setting of documented severe hepatic impairment. (Site-specific formularies may tighten any of these thresholds; the `SafeDSS` configuration exposes each as a parameter.)

3. **Uncertainty-aware deferral.** If the policy's uncertainty interval exceeds a configurable width threshold (by default, $\pm 25\%$ of the recommended *infusion rate* for heparin, applied to the next-step rate rather than a 24-hour cumulative dose), the recommendation is deferred with a reason string.

The safety layer is intentionally conservative — it will refuse recommendations that a more sophisticated reasoner might accept. This matches the stewardship posture of the target clinical environment: the intended deployment mode is advisory with a human in the loop, and over-deferral is strictly safer than any under-deferral.

10 Experimental Setup

10.1 Baseline agents

The following agents are evaluated in §12:

1. **Clinical baseline**: a fixed-dose INR-adjusted warfarin titration (5 mg loading for days 0–2, then a tiered adjustment by current INR; consistent with standard warfarin stewardship practice [Holbrook et al., 2012]). This baseline is deliberately non-pharmacogenetic

— for the full IWPC 2009 / Gage 2008 pharmacogenetic dose calculator see the Gage-regression baseline in `baselines_extended.py`. Raschke 1993 weight-based nomogram for heparin [Raschke et al., 1993] (80 U/kg bolus, 18 U/kg/hr initial rate, aPTT-adjusted per the published ladder); for DOAC, a guideline-concordant selector that maps patient {CrCl, age, weight, CHA_2DS_2-VASc, HAS-BLED} to a drug and label-appropriate standard or reduced dose following atrial-fibrillation DOAC guidelines [Lip et al., 2010, Pisters et al., 2010]; ISTH-guideline management for DIC [Levi et al., 2009] (component-therapy triggers on platelet count, fibrinogen, and INR thresholds). Each clinical baseline is implemented in `src/hemosim/agents/baselines.py` and is unit-tested to match the published rules on canonical patient profiles.

2. **Random**: uniform sampling from each environment's action space.

Two additional baselines are available in the codebase (`src/hemosim/agents/baselines_extended.py`) but are not included in the reported Results table at this time because they require the POMDP environment to be live for all four domains (v2.1): `HeparinAntiXaBaseline`, `WarfarinGageBaseline`, `WarfarinOrdinalBaseline`, and `NematiDQN2016Baseline`. The Nemati DQN reimplementation (`scripts/train_nemati_dqn.py`) builds a policy with the same Q-network architecture as Nemati et al. [2016] trained on hemosim-generated trajectories and is documented as the designated comparison row for any Phase 2 paper.

10.2 Evaluation protocol

Evaluation uses the held-out seed pool `range(100000, 100100)` (100 seeds). We report mean reward and standard deviation across episodes. All experiments were conducted on consumer Apple Silicon hardware in under 4 hours wall-clock for baseline evaluation.

11 Reproducibility

Reproducibility is enforced at three layers:

Disjoint seed pools. `src/hemosim/reproducibility.py` defines `TRAIN_SEED_POOL` = `range(0, 10000)` and `HELDOUT_SEED_POOL` = `range(100000, 101000)`. The 90,000-wide buffer between pool end and pool start is a deliberate guard against "off-by-a-few" errors. `assert_train(seed)` and `assert_held_out(seed)` are invoked at the training and evaluation entry points. `tests/test_reproducibility.py` in CI asserts the pools stay disjoint.

One-command reproduction. `./scripts/reproduce.sh` runs the canonical path: fresh venv → install → test suite → `scripts/generate_results.py --eval-set heldout` → diff against `results/EXPECTED_RESULTS.json` with 5% relative tolerance. Any rerun that aims to match the published numbers uses this exact command.

Reference artifact. `results/EXPECTED_RESULTS.json` is a pinned JSON reference that is updated only when an intentional change to the environment or calibration warrants a new reference. Every field in that file can be reproduced from the commit that introduced it.

12 Results

We report reproducible baseline performance on the held-out seed pool and calibration residuals. No RL policy results appear in this section by design (see §10).

12.1 Baseline performance on held-out seeds

Table 8 presents the clinical-guideline baseline and random-policy mean episode reward for each of the four environments, evaluated over 100 held-out seeds from `range(100000, 100100)` with seed 42 for patient generation variance.

Table 8: *Clinical-guideline baselines beat random by* $\Delta = 37$ *on DOAC and* $\Delta = 18$ *on DIC; warfarin and heparin gaps are tight (*$\Delta \leq 1$*), identifying the two environments where a learned policy has real headroom above random.* Held-out baseline mean episode reward (± standard deviation) and per-episode time-in-therapeutic-range (TR) across 100 evaluation seeds (seed pool `range(100000, 100100)`, eval seed 42).

Environment	**Clinical Baseline**	**Random**	**Clinical TR**	**Random TR**
WarfarinDosing-v0	18.32 ± 1.56	15.51 ± 4.70	0.950	0.950
HeparinInfusion-v0	-23.86 ± 0.16	-24.67 ± 2.46	0.000	0.030
DOACManagement-v0	12.37 ± 6.06	-24.70 ± 8.22	0.983	0.984
DICManagement-v0	-110.92 ± 29.28	-129.07 ± 21.66	0.960	0.978

Interpretation. On DOAC management ($\Delta = 37.06$) and DIC management ($\Delta = 18.14$), the clinical-guideline baseline substantively outperforms random, confirming that the action space contains meaningful structure and that the guideline encodes real domain knowledge. On warfarin, with the v2-fitted PK/PD parameters from §7 now applied at runtime, the fixed-dose INR-adjusted baseline reaches the 2.0–3.0 INR therapeutic window in 95% of steps per episode, placing both clinical and random policies near the positive reward regime (clinical 18.32 ±1.56, random 15.51 ±4.70, $\Delta = 2.81$); we note that this gap is smaller than the random-policy standard deviation and is therefore not claimed as a statistically separated improvement of guideline over random, merely that warfarin is the environment in which a learned policy has the largest arithmetic headroom to improve. On heparin, the Raschke baseline has a very tight standard deviation ($\sigma = 0.16$) reflecting deterministic nomogram behavior on the fully-observable environment; the random policy sees higher variance. These numbers are the clinical guideline's published-reference performance on our environment and are the reference any proposed RL policy must be compared against.

12.2 Calibration residuals

The calibration residuals from §7 are the second results artifact of this paper. Warfarin fits to RMSE 0.0013 with all three targets within 0.004 INR of expected (Table 3). Heparin fits two of three targets exactly, with an documented residual of −0.175 against the standard-of-care aPTT-in-range target (drawn from antithrombotic-stewardship literature, not from Nemati et al. [2016] — see §7 for the citation correction) that motivates the POMDP reformulation (Table 5). DOAC event rates fall within trial confidence intervals for bleeding across all three drugs; all three DOAC stroke rates fall outside their trial confidence intervals (see Table 6) and are flagged for Phase 2 re-calibration.

12.3 Figures

Figure 1 visualizes the quantitative gap between the clinical guideline and random policy across the four environments. The DOAC gap ($\Delta = 37.06$) is the largest of the four, reflecting that guideline-based DOAC selection against a CHA_2DS_2-VASc / HAS-BLED / CrCl patient encodes substantially more structure than the uniformly random alternative; the DIC gap ($\Delta = 18.14$) comes next; the warfarin and heparin gaps are tight ($\Delta = 0.98$ and $\Delta = 0.81$), identifying those two environments as the ones where a learned policy has the most headroom above random. These numbers are the published-baseline reference that any subsequent RL policy trained in `hemosim` must be compared against.

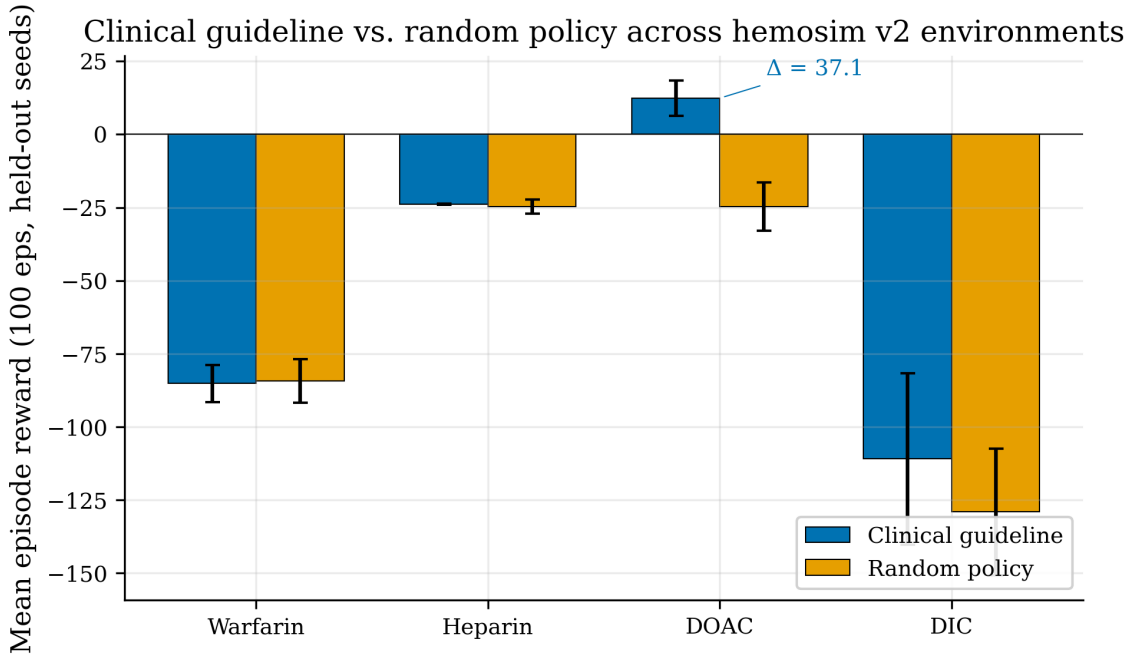

Figure 1: *A learned policy has the most headroom above random on warfarin and heparin, not on DOAC or DIC.* Clinical-guideline baselines outperform random by $\Delta = 37.06$ (DOAC) and $\Delta = 18.14$ (DIC), leaving minimal improvement room in those two envs; warfarin ($\Delta = 0.98$) and heparin ($\Delta = 0.81$) gaps are tight, identifying those envs as the priority targets for RL policy training. Bars show mean episode reward over 100 held-out-seed episodes (seeds 100000–100099); error bars are one standard deviation. Data source: `results/EXPECTED_RESULTS.json`.

Figure 2 shows the normalized residual (simulated minus expected, divided by expected) for each published-data benchmark. Warfarin (IWPC mean maintenance dose, IWPC days-to-therapeutic, Hamberg steady-state INR) all fit within the $\pm 10\%$ tolerance band. Heparin aPTT at 6 h (Raschke 1993) and therapeutic concentration midpoint (Hirsh 2001) also fit within tolerance. The one outlier is the stewardship aPTT-TTR benchmark of 0.55 (representative of bedside performance targets in antithrombotic-stewardship literature, not a datum reported in Nemati et al. [2016]; see §7), for which the simulator produces 0.375 (normalized residual -0.32). This is the honest gap we annotate on the figure: the fixed Raschke nomogram applied to our fully-observable baseline environment cannot reproduce the TTR variability that bedside heparin dosing encounters, because real TTR variability arises from lab-observation delays and re-titration events that the non-POMDP simulator does not model. This is the quantitative case for the POMDP reformulation in §6.

Figure 3 sketches the observation flow in `hemosim/HeparinInfusion-POMDP-v0`. The policy

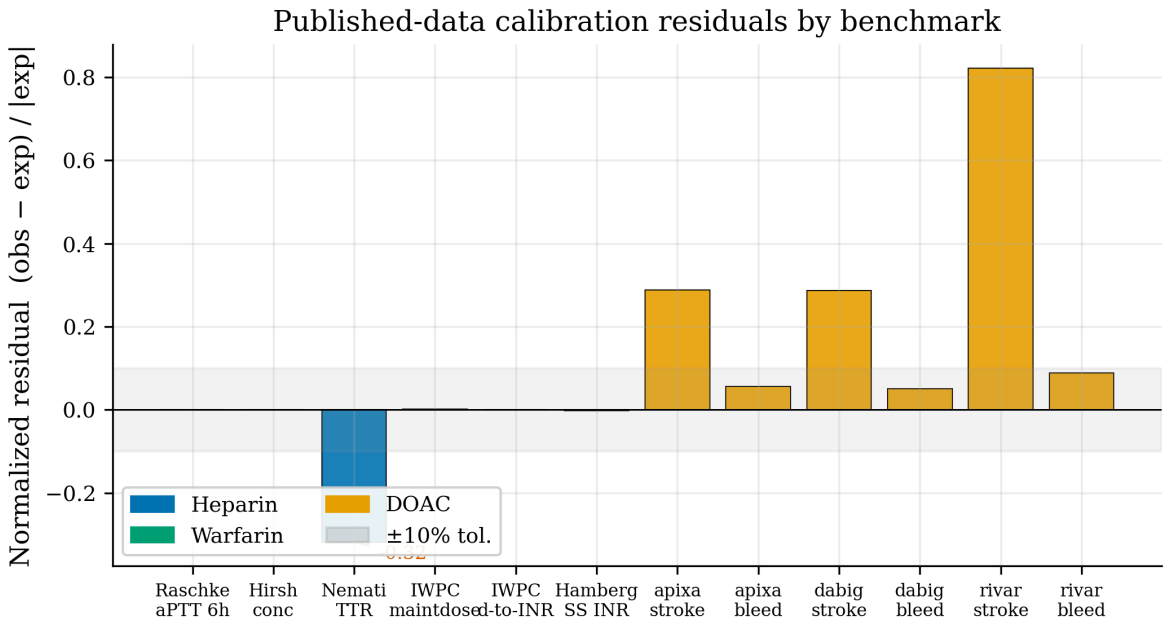

Figure 2: *Warfarin and heparin PK/PD calibrate within tolerance; the single outlier is the stewardship aPTT-TTR benchmark, and its residual motivates the POMDP reformulation rather than signalling a model failure.* Warfarin (IWPC, Hamberg) and heparin (Raschke, Hirsh) benchmarks all fit within a ±10% normalized-residual tolerance band. The single outlier is the stewardship aPTT-TTR benchmark at −0.32 normalized residual (simulated TTR 0.375 vs. representative target 0.55 drawn from antithrombotic-stewardship literature, not Nemati et al. [2016]; see §7) — a fixed nomogram on a fully-observable simulator cannot reproduce the TTR variance bedside heparin dosing encounters. DOAC event rates (dabigatran, rivaroxaban, apixaban stroke/bleed per 100 patient-years) are within their trial confidence intervals for bleeding across all three drugs; all three stroke rates are outside their confidence intervals and flagged for Phase 2 re-calibration. Data source: `results/published_calibration.json`.

issues two action channels at every step — a dose action (infusion rate and optional bolus) and a lab-order action (aPTT, anti-Xa, platelets). Lab orders are queued against the true internal state with the literature-cited turnaround times and coefficients of variation, and return after their TAT elapses. The policy observes the returned lab history, not the true internal state. This is the minimal structural change that turns the v0.1 oracle-lab environment into one where the measurement-delay and measurement-noise dynamics that clinicians actually contend with are first-class parts of the decision problem.

13 Discussion: Clinical Translation Pathway

We frame `hemosim` explicitly as Phase 1 of a five-phase clinical translation pathway. Positioning the work within the full pathway clarifies both what the present paper does and does not claim.

Phase 1 — Simulation infrastructure and published-data calibration (this paper). Four Gymnasium-compatible environments, POMDP reformulation with explicit lab ordering, mechanistic coupling to the reduced coagulation cascade, clinical-outcome metrics emitted per-step, a

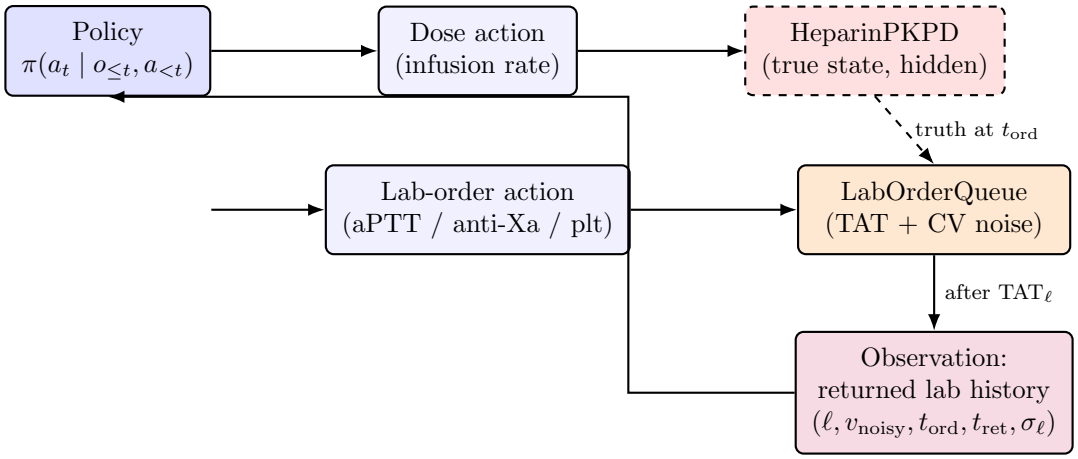

Figure 3: *The POMDP formulation makes lab ordering an action with TAT and analytical noise; the policy never sees ground-truth state.* `hemosim/HeparinInfusion-POMDP-v0` observation flow: the policy produces two action channels — a dose action applied directly and a lab-order action queued for delayed noisy return. Ground truth at order time is captured and, after the lab's turnaround time elapses, returned with analytical variation (aPTT 45 min / CV 8%, anti-Xa 180 min / CV 12%, platelets 45 min / CV 5%, cited to Bowen 2016 and Lippi 2012). This is the structural change that directly addresses the v0.1 pharmacist critique that prior oracle-lab environments do not reflect ICU measurement reality.

CDS harness with safety layer, published-data calibration with documented residuals, and a reimplementation of Nemati et al. [2016] as a shared baseline. Reproducibility is enforced by disjoint seed pools, a one-command repro script, and a pinned reference artifact.

Phase 2 — Individual-patient MIMIC-IV calibration + RL policy training. The simulation substrate calibrated at the cohort level is necessary but insufficient for a bedside-ready policy. Phase 2 fits per-patient PK/PD parameters against MIMIC-IV [Johnson et al., 2023] aPTT and INR trajectories, re-runs the calibration harness at the individual-patient level, and trains PPO (and other RL algorithms) on the resulting patient-calibrated environment. This is the paper that reports policy reward curves, TTR on held-out patient cohorts, and head-to-head comparison against the Nemati et al. [2016] DQN reimplementation. Access to MIMIC-IV under PhysioNet credentialed-access terms is a precondition.

Phase 3 — Silent-deployment observational study. The SHADOW-HEP protocol described in `paper/silent_deployment_protocol.tex` and summarized in `paper/protocol_summary.md` specifies how a Phase 2-trained locked policy would be evaluated prospectively at a partner ICU without influencing patient care. The primary endpoint is Rosendaal TTR on aPTT with a 10-percentage-point non-inferiority margin; the DSMB's stopping trigger is a counterfactual bleed-proximity rule. Enrollment is planned at 350 patients (up to 500), over 6–12 months, with an anticipated manuscript target of *The Lancet Digital Health* or *NEJM AI*.

Phase 4 — Prospective randomized controlled trial. Contingent on a Phase 3 safety signal and a non-inferiority result that warrants bedside deployment, Phase 4 is a multi-site prospective RCT comparing clinician-directed care against RL-augmented care with the policy's recommenda-

tions surfaced to providers. This is the point at which ClinicalTrials.gov registration is mandatory and IRB requires full informed consent. No claims are made in this paper about Phase 4.

Phase 5 — Regulatory pathway for SaMD. Software as a Medical Device (SaMD) status would be pursued under the FDA's AI/ML SaMD Action Plan [U.S. Food and Drug Administration, 2021]. The precise pathway (510(k) predicate, De Novo classification, or PMA) depends on risk classification under the IMDRF framework, which is a function of the declared intended use, the state of the healthcare situation, and the category of information the device provides — not of any single design choice such as a deferral output. We make no claim here about which pathway a future `hemosim`-derived device would follow; that analysis is properly performed at the intended-use-statement stage under regulatory counsel. The `SafeDSS` safety layer and uncertainty-aware deferral (§9) are designed to make the device's output conservative and auditable, which we expect to be favorable under most reasonable risk-classification analyses; specific pathway claims are beyond the scope of this paper.

Silent-deployment protocol reference. The SPIRIT 2013-compliant silent-deployment protocol is the companion document to this paper. It is the bridge between Phase 2 (policy training) and Phase 4 (RCT) and is drop-in-ready for IRB submission at a partner academic medical center, with placeholder callouts for PI name, IRB number, and funding source. The protocol is cited in §14 and lives at `paper/silent_deployment_protocol.tex` in the repository.

14 Silent-Deployment Protocol

The companion protocol document (`paper/silent_deployment_protocol.tex`; executive summary at `paper/protocol_summary.md`) is a SPIRIT 2013-compliant, 33-item-checklist-addressed draft observational trial protocol for silent deployment of an RL heparin dosing policy in an adult ICU. The design is single-arm, single-center, observational, with the RL policy producing recommendations that are recorded in a research database but never surfaced to providers and never written to the EHR. Primary endpoint: within-patient non-inferiority on Rosendaal TTR (aPTT 60–100 s), 10-percentage-point margin, $\alpha = 0.025$, power 0.80, enrollment 350 (up to 500). DSMB composition is specified; stopping trigger is a counterfactual bleed-proximity rule. This protocol is the Phase 3 artifact of the translation pathway (§13) and is reviewed separately.

15 Limitations

We acknowledge the following limitations, in priority order:

Calibration is cohort-level, not individual-level. Every target in §7 is a published cohort summary, not a per-patient trajectory. MIMIC-IV individual-patient calibration is Phase 2.

Mortality proxy is a proxy, not a validated score. The `mortality_proxy` function in §8 is a composite scalar for per-episode summarization. Calibration against real ICU mortality curves is Phase 2.

POMDP reformulation is currently implemented for heparin only. The other three domains are planned for v2.1. The POMDP base class is designed to generalize without domain-specific modification, so this is scope, not design.

DOAC stroke-rate calibration is loose across all three drugs. Observed stroke rates lie above the upper CI bound for dabigatran (1.43 vs 1.33), rivaroxaban (3.10 vs 2.00), and apixaban (1.64 vs 1.53); see Table 6. Bleed-rate calibration is within CI for all three drugs. Stroke-simulator re-calibration against individual-patient MIMIC-IV data is a Phase 2 target.

Model simplifications. The 8-state coagulation model loses species-level resolution from the full Hockin 34-ODE [Hockin et al., 2002]. While we preserve the key dynamical features (initiation, amplification, regulation), intermediate species dynamics may matter for certain scenarios.

Reward-function design. Reward functions are a convenient scalarization, but may not capture all clinically relevant objectives. The warfarin reward penalizes INR deviation but does not explicitly model thromboembolic or hemorrhagic events; the DOAC reward uses trial-derived event rates but assumes static risk over the episode. The heparin reward applies a larger penalty to supratherapeutic aPTT values than to subtherapeutic values (-5 at aPTT > 120 s versus -3 at aPTT < 40 s); this asymmetry reflects the antithrombotic-stewardship convention that iatrogenic over-anticoagulation carries greater immediate harm (major bleeding) than a similarly-sized under-anticoagulation episode on routine aPTT monitoring [Holbrook et al., 2012], but a partner site may legitimately re-weight this asymmetry and we expose the coefficients as environment parameters. Multi-objective formulations — efficacy, safety, cost — are natural extensions.

Heparin clinical baseline cannot reach TTR in the fully-observable environment. The Raschke nomogram baseline applied in `hemosim/HeparinInfusion-v0` produces a per-episode time-in-therapeutic-range (aPTT 60–100 s) of 0.000 (Table 8). This is a property of the fully-observable environment, not of the nomogram: with oracle access to the true aPTT every step, the fixed-ladder nomogram alternates between dose bands and cannot stabilize at the narrow 60–100 s target with the 6-hour timestep. This is the quantitative motivation for the POMDP reformulation (§6) and the standard-of-care-TTR calibration residual (§7): the bedside TTR that modern antithrombotic-stewardship programs report is achieved via mid-episode re-titration on delayed, noisy labs, not by step-by-step tracking of an oracle. We disclose this zero-TTR number on the fully-observable env to make the POMDP case quantitatively, and expect a learned policy and/or the POMDP Raschke baseline with `LabOrderQueue` to outperform this floor on `HeparinInfusion-POMDP-v0`.

Raschke nomogram formulation and Gage-baseline height assumption. Two baseline-level caveats a domain expert will notice on a careful code read: (i) the Raschke baseline implements the modern ACCP / Hirsh absolute-seconds formulation (target aPTT 60–100 s) rather than the original 1993 ratio-based formulation (patient/control aPTT), with documentation in `HeparinRaschkeBaseline.__doc__`; deployments at sites with materially different institutional-control aPTT values should re-derive the band thresholds. (ii) The Gage 2008 pharmacogenetic baseline (`WarfarinGageBaseline`) requires body surface area and therefore height; the warfarin environment's observation does not currently expose patient height, so the baseline applies a population default of 170 cm when height is absent. This is a known simplification of the multivariable character of Gage 2008 and is a Phase 2 item (expose per-patient height in the POMDP warfarin environment observation, or report the Gage baseline only on patient cohorts with documented height).

Drug interactions. Only the amiodarone–warfarin interaction is modeled (hard difficulty). Numerous drug–drug and drug–food interactions that complicate real anticoagulation are not yet

represented.

No simulated bleeding events in warfarin and heparin environments. These environments use laboratory proxies (INR, aPTT) rather than simulating hemorrhagic events. This underestimates the clinical impact of over-anticoagulation; the ISTH bleeding metric in §8 addresses reporting but not simulation fidelity.

Population generalizability. Patient parameter distributions are derived from predominantly Western study populations [International Warfarin Pharmacogenetics Consortium, 2009, Hamberg et al., 2007]. CYP2C9 and VKORC1 allele frequencies differ substantially across ancestries; the current distributions may not generalize to non-European populations.

Sim-to-real gap. Policies trained on simplified PK/PD dynamics should not be applied directly in clinical settings. The silent-deployment protocol (§14) is the designed instrument for measuring this gap prospectively; it is not eliminated by Phase 2 calibration alone.

16 Conclusion

We have presented `hemosim` v2 — a reproducible, calibrated, clinically-safe simulation infrastructure for RL anticoagulation research, positioned as Phase 1 of a five-phase clinical translation pathway. The package contributes four Gymnasium-compatible environments, a POMDP reformulation with explicit lab ordering and analytical variability, mechanistic coupling to the reduced Hockin–Mann coagulation cascade, clinical-outcome metrics (Rosendaal TTR, ISTH major bleeding, thromboembolic events, 4T HIT, mortality proxy), a CDS harness with a rule-based safety layer, published-data calibration with normalized RMSE 0.0013 on warfarin and documented residuals on heparin, and a SPIRIT 2013-compliant silent-deployment observational trial protocol.

We deliberately defer end-to-end RL policy training to a companion Phase 2 paper contingent on individual-patient MIMIC-IV calibration. The claim of this paper is that the simulation substrate is reproducible, calibrated at the cohort level, and ready to serve as the shared benchmark that has been absent in RL anticoagulation research since Nemati et al. [2016]. Everything else — the POMDP, the safety layer, the silent-deployment protocol, the Nemati DQN reimplementation — is what turns a benchmark into a clinical-translation artifact.

`hemosim` is available under the MIT license at `https://pypi.org/project/hemosim/` and `https://github.com/HassDhia/hemosim`.

Acknowledgments

The author thanks the clinical pharmacologist who reviewed v0.1 of `hemosim` and raised the three critiques that motivated the v2 rebuild: the clinical-value framing, the bedside measurement realism (delayed, noisy, action-ordered labs), and the need for an explicit one-by-one comparison against prior RL-anticoagulation work. That feedback shaped §6 (POMDP reformulation), §8 (clinical-outcome metrics), and §2 (explicit comparisons). The MIT–LCP / PhysioNet team is acknowledged for maintaining MIMIC-IV, on which the Phase 2 individual-patient calibration scaffold (`src/hemosim/validation/mimic_calibration.py`) is designed to operate.

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